Harrison Ossias

Toronto, ON | linkedin.com/in/harrison-ossias

Summary

- Robotics and IoT software lead with 6+ years of experience building embedded, telemetry, and cloud systems, from device firmware to real-time streaming pipelines and AI-assisted tooling
- Led engineering teams of 3 at two companies, owning fleet platforms across 21 hospital sites and 14 utility deployments
- Delivers software in regulated environments across healthcare, utilities, and financial services, including a passed first FDA audit, HIPAA and PIPEDA data residency, and GDPR-compliant systems
- Dual citizen authorized to work in Canada and the United States

Professional Experience

Full-Stack IoT & Telemetry Developer, Controls Team Lead **10/2025 – Present**
Able Innovations Inc. – Toronto, ON

- Lead a team of 3 developers owning the edge-to-cloud software stack for a 21-site medical robot fleet
- Shipped end-to-end robot telemetry streaming: a Rust/Python on-device uplink (Zenoh to MQTT to AWS IoT Core) and a Flink/Python processing pipeline on Confluent Kafka, cutting average latency from 30+ seconds to under 2 seconds with zero data loss
- Built a web application delivering real-time analytics and predictive alerting to robot operators and field technicians, saving ~$40K in servicing costs in its first year
- Designed two AWS deployment models, a multi-region fleet platform and isolated per-tenant clusters built to FDA requirements, and led engineering readiness for the company's first FDA audit
- Collaborated with quality assurance to build custom AI tooling on MCP servers, including an in-app assistant that answers fleet analytics questions in plain English, saving ~10 engineering hours a week, and a bot that triages bug reports into prioritized Jira tickets with proposed fixes
- Automated AWS IoT Core device onboarding, including over-the-air (OTA) updates, certificate provisioning, and device-serial management
- Maintain 5 production code bases with test suites validating hardware-to-cloud data flow across embedded (STM32, FreeRTOS), backend, and frontend software

Software & Firmware Engineer (Contract) **01/2025 – 08/2025**
Blue Ocean Technologies Inc. – Toronto, ON

- Ported power-management firmware across ARM architectures (NXP Kinetis Cortex-M4 to i.MX RT Cortex-M7), adapting peripheral and timing-critical code with no change in device behavior
- Converted the flagship Qt application to a cross-platform React and Electron app with a Django and PostgreSQL backend in 3 months
- Integrated and tested Protocol Buffer APIs and UART, SPI, and I2C interfaces for sensor and peripheral communication

Lead Full-Stack Software Engineer **04/2022 – 04/2024**
Hyperion Sensors Inc. – Toronto, ON

- Led a team of 3 building an IoT telemetry SaaS platform across 14 production deployments for 6 Ontario electricity distributors, including Elexicon and Alectra, owning the backend, frontend, and AWS infrastructure
- Re-architected a legacy Tomcat monolith into Kubernetes microservices (Go, NestJS, Django, GraphQL) with MQTT and Kafka streaming, cutting data latency from over a minute to under a second

- Developed C++ data-acquisition software for Linux field devices, processing fiber-optic sensor data at the edge and streaming it over MQTT, with automated OTA updates and release scheduling across the fleet

Associate Software Engineer (Co-op) **05/2021 – 12/2021**
Broadridge – Toronto, ON

- Built a client-release scheduling and system monitoring tool for AWS clusters, integrating with team-wide CI/CD pipelines to streamline deployments using Django, PostgreSQL, and Next.js
- Designed normalized SQL schemas with GDPR-compliant data handling, including data minimization, access controls, and retention policies
- Delivered hotfixes and client patches in an Agile team, taking part in daily stand-ups and code reviews

DevOps/SRE Cloud Developer (Co-op) **05/2020 – 12/2020**
IBM – Ottawa / Toronto, ON

- Served as on-call site reliability engineer (SRE) and DevOps developer for IBM Planning Analytics on Demand across 25 clients, including IKEA, PepsiCo, and EY
- Automated performance monitoring with Python and PowerShell across major client environments, reducing authentication-related downtime by ~2.5%
- Managed 60+ production and development environments across Active Directory and bare-metal Linux using Ansible Tower, enforcing GDPR-aligned access controls for EU client environments

Data Analyst, tenthree labs (Co-op) **01/2020 – 04/2020**
Ernst & Young (EY) – Toronto, ON

- Built support applications in C++ and Python to handle financial statement cleansing, validation, and year-to-year rollover for 39 corporate audits
- Deployed cleansing pipelines in Alteryx using PostgreSQL and Microsoft SQL Server as data stores

Technical Skills

- **Languages:** Python, TypeScript/JavaScript, C++, Rust, Go, SQL
- **Embedded & Robotics:** STM32, ARM Cortex-M4/M7, NXP Kinetis/i.MX RT, FreeRTOS, Zephyr, ROS 2, Zenoh
- **Cloud & DevOps:** AWS (IoT Core, EC2, ECS, EKS, Lambda, RDS, S3, IAM, Cognito), GCP, IBM Cloud, Kubernetes, Ansible Tower, CI/CD, OTA Updates, Linux
- **Data & Streaming:** Apache Kafka (Kafka Streams, Confluent), Apache Flink, PostgreSQL, SQLite, Redis
- **Protocols & APIs:** MQTT, gRPC, Protocol Buffers, GraphQL, REST, CAN, UART, SPI, I2C
- **Web:** React, Next.js, Node.js, NestJS, Express.js, Django, Electron
- **AI Tooling:** Claude, Model Context Protocol (MCP)

Education

MEng, Mechanical & Industrial Engineering (Robotics) **01/2025 – 08/2026**
University of Toronto – Toronto, ON

- Specialized in robotic perception and physical AI systems, using ROS 2, Isaac Sim, and PyTorch
- Completed research in telepresence robotics and medical robot adoption across Canada

BTech, Automation Engineering (Smart Systems & IoT Engineering) **09/2017 – 05/2022**

McMaster University – Hamilton, ON